

1. Administrative & Study Identification

Full Study Title: A Phase 2, randomized, open-label, controlled study to evaluate the efficacy and safety of rapcabtagene autoleucel versus comparator in participants with severe refractory idiopathic inflammatory myopathies (IIM) **Short Title/ Acronym:** Phase 2 Study of Rapcabtagene Autoleucel in Myositis **Protocol Number:** CYTB323L12201 **NCT Number:** NCT06665256 **Sponsor Name:** Novartis **Investigational Product:** Rapcabtagene autoleucel (YTB323) / Autologous CD19 CAR-T **Phase of Development:** Phase 2 **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of rapcabtagene autoleucel (administered once following lymphodepletion) versus an active comparator option in participants with severe refractory idiopathic inflammatory myopathies (IIM). **Secondary Objectives:** To assess the cellular kinetics, pharmacokinetics, long-term safety, biomarkers, and B-cell depletion/recovery profiles associated with the "immune reset" mechanism, as well as specific glucocorticoid dose reductions and improvements in respiratory and total improvement scores.

2.2 Background & Rationale To address the gap in effective treatments for patients with severe refractory myositis/idiopathic inflammatory myopathies (IIM) who have an inadequate response to standard-of-care therapies (e.g., corticosteroids and immunosuppressants). Rapcabtagene autoleucel uses CD19-directed CAR-T cell therapy to target and deplete B-cells, offering a highly targeted alternative to reprogram the patient's immune system and induce a potentially durable "immune reset".

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's choice of treatment) **Blinding:** Open-label, Assessor-blinded **Randomization:** Randomized. The study comprises two cohorts: a lead-in cohort enrolling participants to receive rapcabtagene autoleucel, and a randomized cohort with participants receiving either rapcabtagene autoleucel or a comparator option.

3.2 Planned Enrollment & Duration Target **\$N\$:** 123 participants
Number of Sites: Approximately 54 locations internationally
Estimated Study Start: 17-Dec-2024 **Study Duration:** Two-year primary study period, followed by a separate Long-Term Follow-Up (LTFU) period lasting up to 15 years after rapcabtagene autoleucel infusion.

4. Subject Selection (Eligibility Criteria)

4.1 Key Inclusion Criteria 1. Men and women, aged >18 and ≤ 65 years. 2. Diagnosis of probable or definite myositis according to American College of Rheumatology/European League Against Rheumatism 2017 (ACR/EULAR 2017) criteria. 3. Participants who had inadequate response to prior therapy. 4. Diagnosed with active disease. 5. Participant must meet criteria for severe myositis.

4.2 Key Exclusion Criteria 1. Any condition during Screening that could prevent a complete washout of medications or could otherwise make the participant ineligible for anti-CD19 CAR-T therapy and further participation in the study. 2. BMI at Screening of ≤ 18.5 or ≥ 35 kg/m2. 3. Severe muscle damage at Screening. 4. Inadequate organ function. 5. Hypersensitivity and/or contraindications to any product (including its ingredients) to be given to the participant as per the study protocol. 6. Other inflammatory and non-inflammatory myopathies. 7. Any medical conditions that are not related to IIM that would jeopardize the ability of the participant to tolerate CD19 CAR-T cell therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single infusion of rapcabtagene autoleucel (YTB323) **Route of Administration:** Intravenous (IV) infusion (following lymphodepletion conditioning) **Duration of Treatment:** Single infusion, followed by intensive monitoring and an extensive post-treatment follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for myositis symptoms as defined per protocol. **Prohibited:** Concurrent experimental therapies, live vaccines, and any targeted immunosuppressants/biologics that conflict with CAR-T cell expansion and engraftment without appropriate washout.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent, Eligibility Assessment, Medication Washout, Leukapheresis

Baseline | Day 0 | Lymphodepletion, Single CAR-T Infusion (Rapcabtagene autoleucel), Safety Audit | | **Interim** | Post-Infusion | Safety Labs (CRS/ICANS monitoring), Efficacy Assessment, Cellular Kinetics | | **End of Study (EOS)** | Year 2 | Final Efficacy/Safety Assessment; transition to a separate 15-year Long-Term Follow-Up (LTFU) protocol |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Proportion of participants achieving at least moderate improvement in Total Improvement Score (TIS) at Week 52:** The percentage of participants with a TIS of at least 40 at the 52nd week after the start of the study, corresponding to at least moderate improvement.

7.2 Secondary Outcomes 1. **Adjusted annual cumulative glucocorticoid dose up to Week 52.** 2. **Change from baseline in percent predicted Forced Vital Capacity (FVC%) at Week 52.** 3. **Proportion of participants achieving major improvement in TIS at Week 52.**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring for CAR-T specific toxicities, particularly Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS). **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor/Regulatory bodies. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee established to review safety data throughout the lead-in and randomized cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy assessments; Safety Set for all participants who receive lymphodepletion/CAR-T infusion. **Sample Size Justification:** $N = 123$ participants appropriately powered to evaluate safety and determine the efficacy of rapcabtagene autoleucel against the active comparator arm. **Handling of Missing Data:** Standard imputation methods as predefined in the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during the peri-infusion window, transitioning to remote/routine visits during the long-term follow-up phase. **Audit Trail:** Robust eCRF

locking and data clarification mechanisms maintained by Novartis quality assurance protocols.

11. Study Identifiers & Governance

Primary ID: CYTB323L12201 **Registry Number:** NCT06665256 **Sponsor/Lead Organization:** Novartis **Study Title:** Phase 2 Study of Rapcabtagene Autoleucel in Myositis **Official Regulatory Title:** A Phase 2, randomized, open-label, controlled study to evaluate the efficacy and safety of rapcabtagene autoleucel versus comparator in participants with severe refractory idiopathic inflammatory myopathies (IIM)

12. Clinical Framework (Myositis Specific)

Primary Condition: Severe Refractory Idiopathic Inflammatory Myopathies (IIM) **Diagnosis Threshold:** Probable or definite myositis according to ACR/EULAR 2017 criteria **Medical Methodology:** Autologous CD19-directed Chimeric Antigen Receptor T-cell (CAR-T) therapy **Optimization Target:** Targeted B-cell depletion leading to immune system reset and reduced muscle inflammation

13. Study Procedures & Methodology

Management Pathway: Leukapheresis → Cell Manufacturing → Lymphodepleting Chemotherapy → CAR-T Infusion **Education Protocol (HCP):** Specialized site training for the administration of CAR-T therapies and acute management of CRS and neurotoxicity. **Education Protocol (Patient):** Comprehensive onboarding regarding CAR-T cell therapy expectations, hospitalization requirements, and signs of severe adverse events. **Quality Audit Mechanism:** Real-time cellular kinetic tracking and stringent safety monitoring protocols post-infusion.

14. Observation & Follow-Up Schedule

Total Duration: Two-year primary study follow-up. After end of study (EOS), participants who received rapcabtagene autoleucel infusion will enter a Long-Term Follow-Up (LTFU) period lasting up to 15 years after the infusion. **Onsite Visit Cadence:** Intensive daily/weekly visits for the first 28 days post-infusion; progressively spaced to quarterly/annual visits thereafter. **Remote Monitoring Frequency:** Supplemental telemedicine safety check-ins as defined by standard CAR-T post-discharge protocols. **Checklist Review Interval:** Regular milestone

reviews leading up to enrollment in the subsequent 15-year LTFU registry.

15. Sign-off & Approval

Role	Name	Signature	Date					
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]					